Practical assignment 1 **Academic year: 2026/2027**

Objective: Analysis of data available in Sentinel-2 SAFE files

1) Download from the Copernicus browser (https://browser.dataspace.copernicus.eu/) two Sentinel-2 products, with the level of processing 2A, corresponding to tile 29TNE. One from the 13th of August 2025 and the other from the 22nd of September 2025 (already used in Exercise Sheet 2). Unzip the obtained files.

2) Analyze the contents of the SAFE folders. You will find information in the webpage https://sentiwiki.copernicus.eu/web/s2-products that will help you to understand the contents of each folder and file. Check in particular the files in folder "GRANULE" and the file "MTD_MSIL2A.xml". Identify the location of the 13 bands. You may add them to QGIS.

3) Identify, at least, the following aspects:

a) What is the radiometric resolution of the bands?
b) Identify the variation of values in each band and what those values represent.
c) Would you need to make any conversion to obtain BOA (Bottom Of Atmosphere) reflectance?
d) Is there another designation now used for BOA reflectance?

4) Is there a file in the SAFE folder that enables the identification of clouds in the images[1]? Identify the product and add it to QGIS.

5) Identify the classes available in the product identified in 4) and display them using the symbology recommended by ESA, available at https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/scene-classification/[2].

6) Create a mask for the 13th of August 2025 image, so that all regions affected by clouds or cloud shadows could be set to "0" and all other pixels to "1"[3]. This file may be used as a mask to process only the parts of the bands that not affected by clouds. How could that be done?

Instructions:

- The work **may** be done in **groups** up to three elements.
- The use of AI (such as ChapGPT, Perplexity or Gemini) to answer any questions you may have in encouraged, but make sure you verify the correctness of the provided answers (and the consistency of answers between chatbots).
- Prepare a **short report** (2 or 3 pages) of the group findings. It should be uploaded in "Infoestudante" until the **14th of October**.
- An **oral discussion** of the group's findings will be made with each group during the class on October **15th of October**.
- **Maximum score**: 1 point (out of 20).

[1] Search for an SCL file.
[2] You can identify the color corresponding to each HTML color code at: https://www.w3schools.com/colors/colors_picker.asp
[3] Use the QGIS tool "Reclassify by table".